

1. What is Inferential Statistics?

- Inferential statistics is the process of using sample data to make conclusions or predictions about a larger population.
It includes methods such as hypothesis testing, confidence intervals, and estimation.

2. What is Hypothesis Testing and its Components?

- Hypothesis testing is a statistical method used to determine whether there is enough evidence to support a claim about a population.
Its main components are Null Hypothesis (H_0), Alternative Hypothesis (H_1), significance level, test statistic, p-value, and decision.

3. Explain Confidence Interval and Critical Value.

- A confidence interval is a range of values within which the true population parameter is expected to lie with a certain level of confidence.
A critical value is a boundary value used to decide whether the test statistic falls in the rejection region.

4. Define P-value.

- A p-value is the probability of obtaining the observed result, or a more extreme result, assuming that the null hypothesis is true.

5. Differentiate Type I and Type II Errors.

- Type I Error: Rejecting the null hypothesis when it is actually true.
Type II Error: Failing to reject the null hypothesis when it is actually false.

6. Brief Descriptions of Z-test, t-test, Chi-square test, and ANOVA.

- Z-test: Used to compare means when the population standard deviation is known or the sample is sufficiently large.
- t-test: Used to compare means, especially when the population standard deviation is unknown and the sample is small.
- Chi-square test: Used to test the relationship or association between categorical variables.
- ANOVA: Used to determine whether there is a significant difference between the means of three or more groups.

7. What is Covariance?

- Covariance measures the direction in which two variables change together.
A positive covariance indicates that they tend to increase together, while a negative covariance indicates that one tends to increase when the other decreases.

8. What is Correlation?

- Correlation measures the strength and direction of the relationship between two variables.

Its value ranges from -1 to +1, where +1 indicates a perfect positive relationship and -1 indicates a perfect negative relationship.